Technical Requirements Document (TRD)

Engineering Blueprint, Database Schema, and Algorithmic Domain Engines

1. System Architecture & Tech Stack

LAYER	TECHNOLOGY	IMPLEMENTATION DETAILS
Frontend SPA	React 18 + Vite 6 + TypeScript 5.7	Modular multi-file layout, custom hooks, and strict type safety
Styling Engine	Tailwind CSS 3.4 + Custom Tokens	4 distinct token CSS files, Google Fonts, and mobile 16px input safeguard
Database	Supabase PostgreSQL 17	11 relational tables with foreign key cascades and Realtime WebSocket events
Graphics Rendering	html-to-image	1200×1200px dedicated DOM container rendered at 2× pixel ratio with skipFonts: true
State & Persistence	React Context + LocalStorage	Synchronous client state with optimistic updates and PostgreSQL reconciliation

2. Supabase PostgreSQL 17 Relational Schema

The system operates on an 11-table relational schema deployed via the Supabase MCP:

TABLE NAME	PRIMARY KEY	FOREIGN KEYS & RELATIONSHIPS	OPERATIONAL PURPOSE
`profiles`	`id (UUID)`	References `auth.users(id)`	Tournament directors, chief scorers, timekeepers
`venues`	`id (UUID)`	1:N with `courts`	Multi-court facilities (Colombo Futsal Hub)
`courts`	`id (UUID)`	References `venues(id)`	Court 1 to 5 with synthetic and turf surfaces
`tournaments`	`id (UUID)`	References `venues(id)`, `profiles(id)`	Core tournament record, status, dates, and budget
`teams`	`id (UUID)`	References `tournaments(id)`	Competing clubs assigned to Group A or B
`players`	`id (UUID)`	References `teams(id)`	Squad members with jersey numbers and positions
`matches`	`id (UUID)`	References `tournaments(id)`, `courts(id)`	Scheduled fixtures, home/away scores, and status
`match_events`	`id (UUID)`	References `matches(id)`, `players(id)`	Goals, assists, yellow/red cards, and kickoff minutes
`court_booking_locks`	`id (UUID)`	References `courts(id)`	Venue maintenance slots and private bookings
`organizer_invites`	`id (UUID)`	References `tournaments(id)`	Delegation tokens for co-organizers
`audit_logs`	`id (UUID)`	References `tournaments(id)`	Immutable audit trail for court changes & walkovers

3. Algorithmic Domain Engines

3.1 Two-Tier Court Conflict Mutex Engine (conflictEngine.ts)

Prevents court schedule overlaps across two critical protection levels:

- **Tier 1 (Intra-Tournament Mutex)**: Validates against concurrent fixtures within the same tournament using interval overlap logic:
  `Overlap = (Court_A == Court_B) && (Start_A < End_B) && (End_A > Start_B)`
- **Tier 2 (Inter-Tournament & Venue Locks)**: Validates against external bookings, court maintenance, and turf repair slots in `court_booking_locks`.
- **Automated Suggestion**: If a clash is detected, the engine scans remaining courts and proposes the first free alternate court.

3.2 FIFA/AMF Futsal Standings Engine (standingsEngine.ts)

Computes real-time rankings with cascading tiebreakers:

1. **Total Points**: `3 * Wins + 1 * Draws`.
2. **Head-to-Head (H2H)**: Evaluates direct match results between squads tied on points.
3. **Goal Difference (GD)**: `Goals For (GF) - Goals Against (GA)`.
4. **Goals For (GF)**: Total goals scored in group play.

3.3 Disciplinary & Forfeit Engine (forfeitEngine.ts)

Enforces official FIFA walkover regulations:

- **Walkover (3-0 Forfeit)**: If a squad fails to report, one click marks the match `WALKOVER`, awards a **3-0 win (+3 PTS, +3 GD)** to the opponent, and assigns 0 PTS and -3 GD to the defaulting squad.
- **Disqualification**: Excludes the team from playoff progression and tags their roster as ineligible.

4. Visual Verification Artifacts

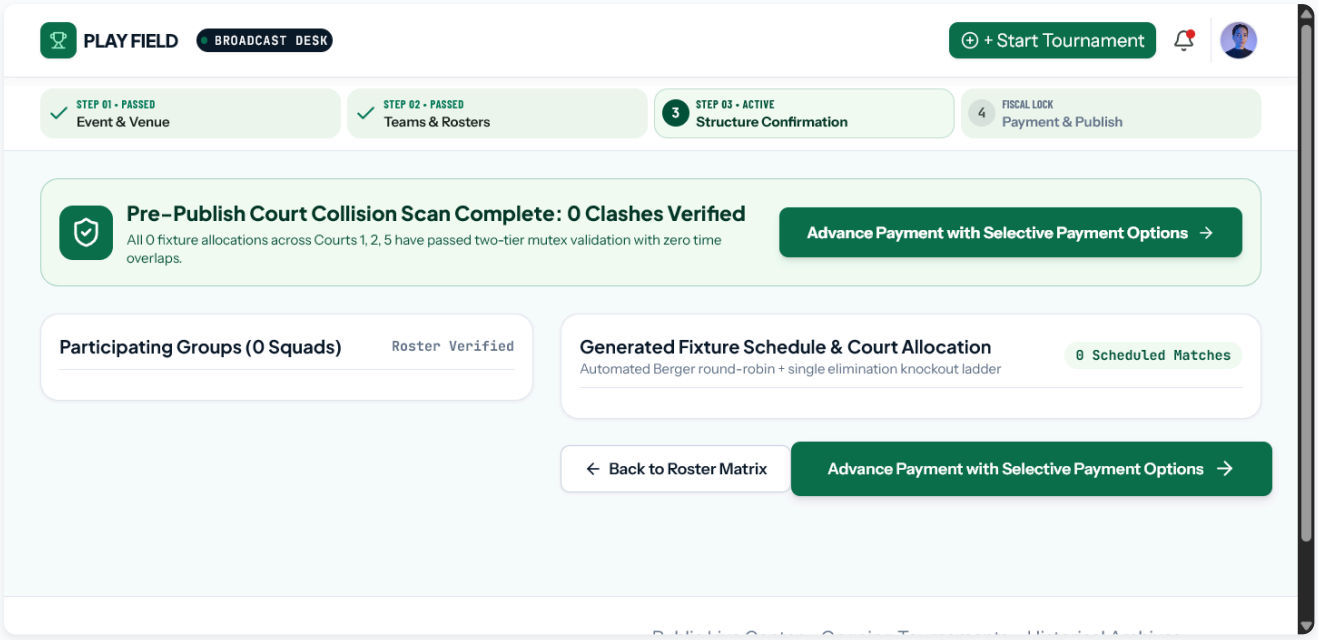


Figure 3: Pre-Publish Court Collision Scan Complete (Step 3 Confirm)

Automated validation confirming 0 clashes across all scheduled fixtures before entering the payment gateway.

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